Wi-Vi (Wireless Vision)

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ABSTRACT

Wi-Fi signals are typically information carriers between a transmitter and receiver. But Wi-Fi can also extend our senses, enabling us to see moving objects through walls and behind closed doors. In particular, we can use such signals to identify the number of people in a closed room and their relative locations. We can also identify simple gestures made behind a wall, and combine a sequence of gestures to communicate messages to a wireless receiver without carrying any transmitting device.

Wi-Vi technology is a wireless device that captures moving objects behind a wall and door. Wi-Vi has the strategic advantage of Wi-Fi to make through wall imaging relatively low cost, low power, low-bandwidth, and accessible to average users.

Wi-Vi is based on the principle of RADAR and SONAR imaging(doppler effect). It's similar to the way radar and sonar work but without the expensive, bulky gear and restricted frequencies that radar requires depending on its own transmitting signal.

Figure 1: Visualisation of Wi-Vi

INTRODUCTION

Can Wi-Fi signals enable us to see through walls? For many years, humans have fantasized about X-ray vision and played with the concept in comic books and sci-fi movies. This poster explores the potential of using Wi-Fi signals and recent advances in MIMO communications to build a device that can capture the motion of humans behind a wall and in closed rooms. Law enforcement personnel can use the device to avoid walking into an ambush, and minimize casualties in standoffs and hostage situations. Emergency responders can use it to see through rubble and collapsed structures. Ordinary users can leverage the device for gaming, intrusion detection, privacy-enhanced monitoring of children and elderly, or personal security when step ping into dark alleys and unknown places. The concept underlying seeing through opaque obstacles is similar to radar and sonar imaging. Specifically, when faced with a non-metallic wall, a fraction of the RF signal would penetrate the wall, reflect off objects and humans, and come back imprinted with a signature of what is inside a closed room. By capturing these reflections, we can image objects behind a wall. Building a device that can capture such reflections, however, is difficult because the signal power after traversing the wall twice is reduced by three to five orders of magnitude.

TRACKING A SINGLE HUMAN

In advanced, through all systems antenna array is employed to trace the human motion. They steer the arrays beam to see the direction of most energy and this direction corresponds to the signals abstraction angle of arrival. By following that angle in time, it is possible to infer however the thing moves in area.

However, Wi-Vi avoids using an antenna array for two reasons: First is in order to obtain a narrow beam that means achieve a good resolution, one needs a large antenna array with many antenna elements. This would result in a bulky and expensive device. Second is, since Wi-Vi eliminates the flash effect using MIMO nulling, adding multiple receive antennas would require nulling the signal at each of them. This requires adding more transmit antennas so the device will become bulkier and more expensive.

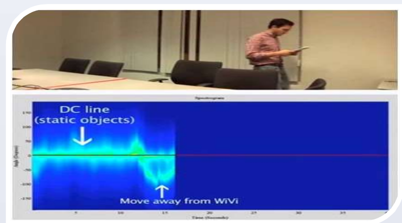

Figure 3: Demonstration of Wi-Vi to let the users 'see' a person moving behind a wall.

TRACKING MULTIPLE HUMANS

With multiple humans, the noise increases significantly. On one hand, each human is not just one object because of different body parts moving in a loosely coupled way and on the other hand, the signal reflected off all of these humans which are correlated in time, hence they all reflect the transmitted signal. The lack of independence between the reflected signals is important. For example, the reflections coming from two humans may combine systematically to dim each other for some period of time.

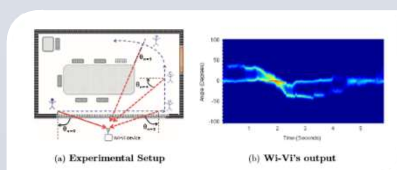

Figure 4: DAS Checking Scheme

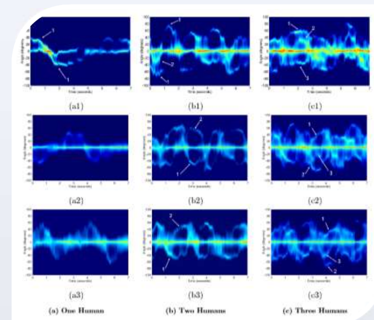

Figure 4: Different Wi-Vi output

ALGORITHM FOR PSEUDO WI-VI NULLING

INITIAL NULLING:
▷ Channel Estimation
Tx ant. 1 sends x; Rx receives y; $\hat{h}_1 \leftarrow y/x$
Tx ant. 2 sends x; Rx receives y; $\hat{h}_2 \leftarrow y/x$
▷ Pre-coding: $p \leftarrow -\hat{h}_1/\hat{h}_2$
POWER BOOSTING:
Tx antennas boost power
Tx ant. 1 transmits x, Tx ant. 2 transmits px concurrently
ITERATIVE NULLING:
$i \leftarrow 0$
repeat
  Rx receives y; $h_{res} \leftarrow y/x$
  if i even **then**
    $\hat{h}_1 \leftarrow h_{res} + \hat{h}_1$
  else
    $\hat{h}_2 \leftarrow \left(1 - \frac{h_{res}}{\hat{h}_1}\right)\hat{h}_2$
  $p \leftarrow -\hat{h}_1/\hat{h}_2$
  Tx antennas transmit concurrently
  $i \leftarrow i + 1$
until Converges

THROUGH WALL GESTURE-BASED COMMUNICATION

Wi-Vi can enable a human who does not carry any wireless device to communicate short messages or commands to a receiver using simple gestures. Wi-Vi represents these try of gestures by „0" bit and „1" bit. These gestures are later composed by human to make messages that are having completely different interpretations. In addition, Wi-Vi will develop by exploitation different existing practices and principles like adding an easy code that may guarantee dependability, or by reserving an exact pattern of „0" and „1"s. At this stage this technology continues to be terribly basic, nevertheless we have a tendency to believe future advancement scan build it a lot of reliable and communicative.

APPLICATIONS

There are some of the applications of Wi-Vi technology described here:

Law enforcement: Law enforcement personal can use the device to avoid walking into an ambush, and minimize causalities in hostage and standoffs situations.
Emergency situations: Emergency responders can use Wi-Vi to see through rubble and collapsed structures.
Smart Sensing: This Wi-Vi technology can be extended to sense motion in different parts of a building and allow automated control of heating or cooling and lighting systems.
Personal Security: Common users can use it for intrusion detection and when stepping into dark alleys and unknown places.
Entertainment: It enables a new dimension for input output devices in gaming which does not affect on occlusion and works in non-line-of-sight. **User Interface Design**: This technology may also be leveraged in the future to enable the controlling household appliances via gestures, and non-invasive monitoring of children and elderly.

CONCLUSION

We discussed Wi-Vi, a wireless technology that uses Wi- Fi signals to detect moving humans behind walls or doors and also in closed rooms. As compared to previous systems, which are targeted for the military, Wi-Vi enables the small cheap see-through-wall devices which operate in the ISM band, rendering them feasible to the general public. Wi-Vi also builds a communication channel between a human behind a wall or in a closed room and device itself, allowing person to communicate directly with Wi-Vi without carrying any of transmitting device. We believe that Wi-Vi has a set of functionality that future Wireless networks will provide. Future Wi-Fi networks will likely expand beyond communications and deliver facilities such as indoor localization, sensing as well as control. Wi-Vi gives evidence of advanced form of Wi- Fi-based sensing and localization by using Wi-Fi to track humans behind wall without carrying any wireless device.

FUTURE WORK

- Wi-Vi could be built into smartphone or a special handheld device.
- Evolution of seeing human through denser building materials and with a longer range.
- High quality images.

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